RESEARCH MILESTONE 01

TERRA WATCH

From change pixels to evidence-grounded descriptions



ILLUSTRATIVE AERIAL VISUAL / NOT A MODEL RESULT

IIT Madras BS / Data Science & AI Lab / October 2026

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01 / THE QUESTION

A change map tells us where.

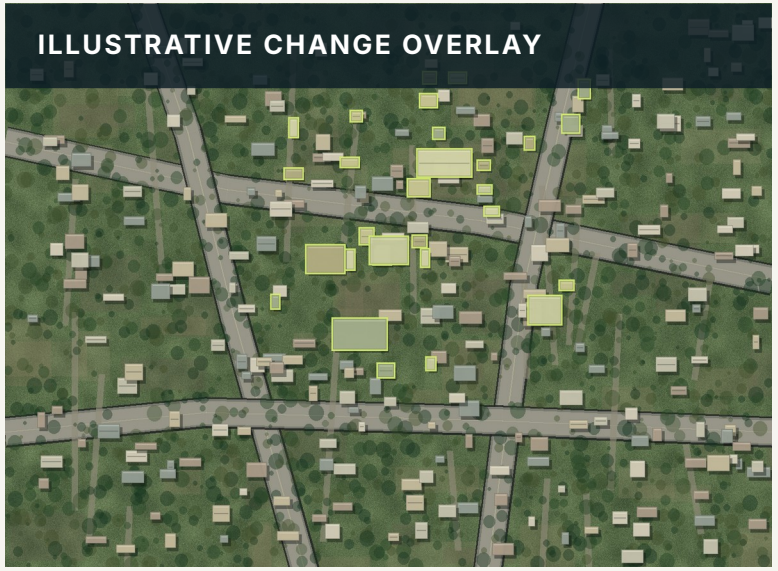
The next question is whether we can explain what happened - without inventing facts.



Two aligned images



A machine-readable mask



A claim we can verify

Concept illustration; not an example from LEVIR and not a trained prediction.

02 / OPERATIONAL GAP

The bottleneck is interpretation.

Current workflows generate alerts. People still spend time deciding which alerts mean something.

01 OBSERVE

Satellite pairs at two dates

Imaging conditions are not identical

→

02 FLAG

Change map highlights pixels

Shadows and seasons trigger noise

→

03 VERIFY

Officer inspects each location

What changed? Is it relevant?

OUR OPPORTUNITY

Turn each flag into a short, localised, evidence-checked description.

03 / REAL WORLD

More imagery than humans can inspect.

Institutional change-detection workflows already exist. Interpreting every alert remains expensive.

4+ TB

new Sentinel-2 products daily

ESA / cited in project report

62

Indian cantonment boards

DGDE monitoring context

2026

Mumbai BMC initiative

reported approval of AI + satellite monitoring

Detection alone does not complete the decision workflow.

Source: Milestone 1 report, Motivation & Practical Relevance (see report references 1, 3, 4).

04 / DELIBERATE FOCUS

Start narrow. Make claims auditable.

Milestone 1 centers on high-resolution built-up change rather than generic Earth monitoring.

IN SCOPE

- Co-registered RGB image pairs (~0.5 m/pixel)
- Buildings, roads and groundwork in built-up areas
- Pixel-level mask and grounded short caption
- Lighting / blur / alignment robustness tests
- Public benchmarks, human audit and local demo

OUT OF SCOPE

- Legal status / ownership / intent
- Causal explanations of why
- Damage or monetary estimates
- Unvalidated Sentinel-2 use
- Nationwide real-time deployment

05 / WHO BENEFITS

Built for evidence-based review.

The benefit is a better starting point for the human verifier, not an automated final decision.

01 PRIMARY

URBAN LOCAL BODIES

Spot newly built structures; review flags on site.

02 PRIMARY

DEFENCE ESTATES

Inspect built-up change around managed land.

03 SECONDARY

PLANNING & RESEARCH

Compare growth and audit what models claim.

Outputs designed for inspection: location · change region · limited, checkable description

06 / RESEARCH LANDSCAPE

Progress has been real. So have the gaps.

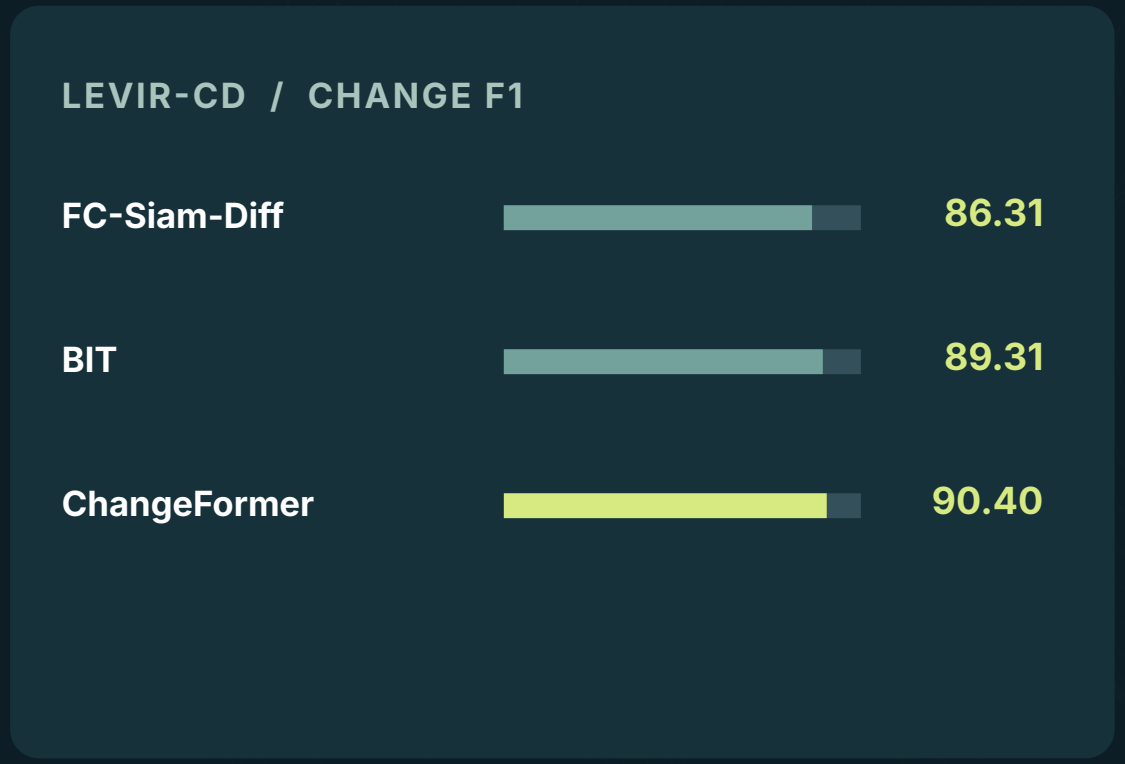
A quick view of six model families discussed in the literature review.

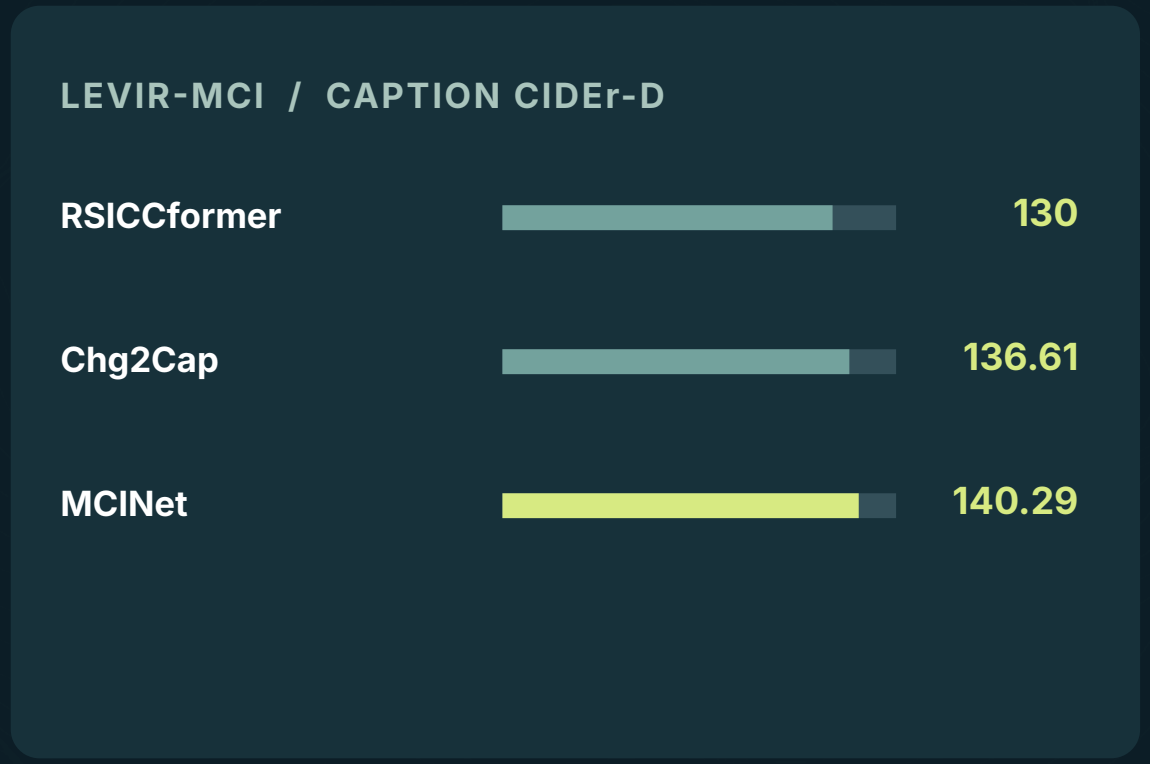
01	**IMAGE DIFFERENCING**	CVA / Otsu	Fast; noise mistaken for change
02	**SIAMESE CONVOLUTION**	FC-Siam / U-Net	Accurate spatial maps; no language
03	**TRANSFORMERS**	BIT / ChangeFormer	Strong benchmark scores; labels required
04	**FOUNDATION MODELS**	AnyChange / RemoteCLIP	Transfer and zero-shot options; mixed accuracy
05	**CHANGE CAPTIONING**	RSICCformer / Chg2Cap	Readable descriptions; overlap metrics
06	**INTEGRATED VISION + LANGUAGE**	Change-Agent / ChangeChat	Multiple tasks; claims need checking

07 / BENCHMARK EVIDENCE

Detection is a strong baseline problem.

Reported scores reproduced from the cited ChangeFormer and Change-Agent tables (not our results).



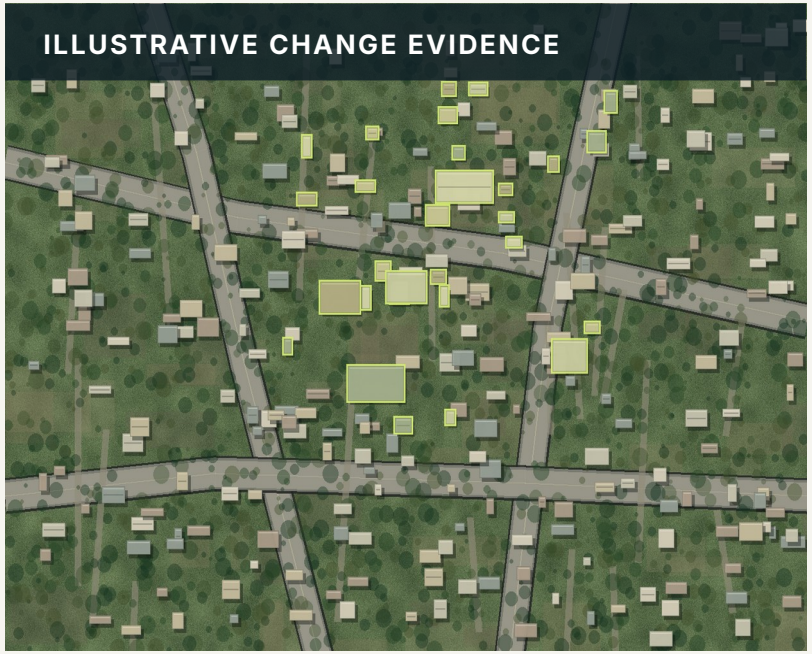


Excellent lexical overlap does not establish factual correctness.

08 / THE GAP

What if the sentence could be checked?

The novelty claim is evidence grounding and evaluation - not inventing a new detector or pairing two models.



UNVERIFIED GENERATION

"Seven buildings were added after illegal clearing."

CHECK INDIVIDUAL CLAIMS

Object class • Count • Relative location • Unsupported inference

"Buildings appear in the central portion of the image."

Example wording is illustrative and does not describe an evaluated model output.

09 / FOUR MEASURABLE OBJECTIVES

Define success before training.

Targets are hypotheses in the Milestone 1 report - not achieved outcomes.

01 DETECT

Change-class F1 / IoU

Compare U-Net, ChangeFormer and classical differencing

02 WITHSTAND

F1 drop under nuisance shift

Lighting, colour, blur, compression, misalignment

03 DESCRIBE

Supported-claim precision

Grounded captions vs image-only generation

04 DELIVER

End-to-end latency

Working local demo and reproducible pipeline

10 / DATASET STRATEGY

One anchor dataset. Several stress tests.

We avoid mixing unrelated benchmarks into one headline score or leaking repeated images across splits.

PRIMARY PAIR

LEVIR-MCI

10,077 paired images with aligned change masks and 5 captions per pair

Detection + language on the same images

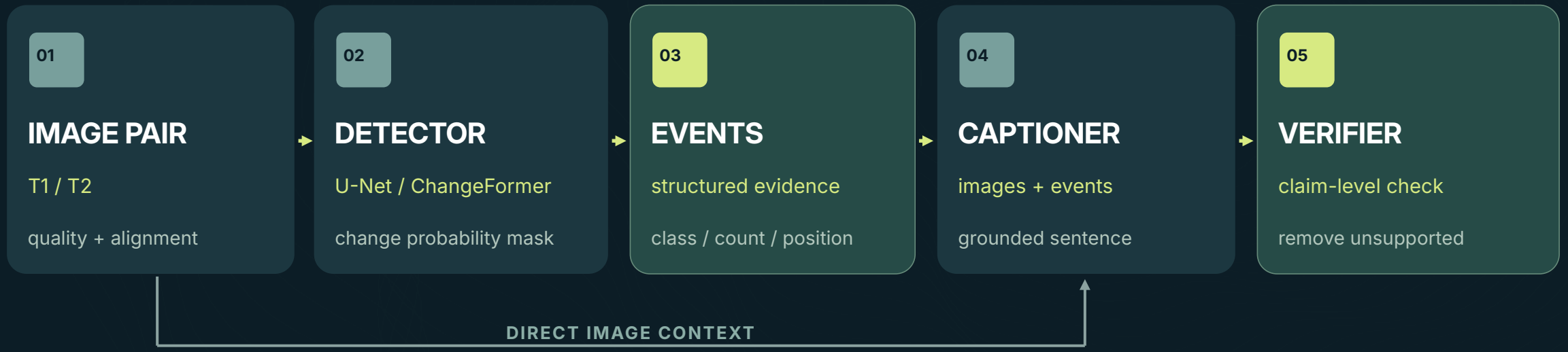
LEVIR-CD building change benchmark	637 pairs
S2Looking side-looking robustness	5,000 pairs
SYSU-CD cross-domain generalisation	20,000 pairs
SECOND optional semantic change	4,662 pairs
ChangeChat-105k optional instruction tuning	~105k Q&A

LEVIR-CC, LEVIR-MCI and ChangeChat share imagery: split and hash before evaluation.

11 / METHOD OVERVIEW

Make the mask part of the evidence.

Modular, replaceable components with explicit claim checking.



M1 proposal only: individual modules and gains are yet to be implemented or evaluated.

12 / ISOLATE THE CONTRIBUTION

Four settings. Same test examples.

Ablations separate the effects of visual input, change evidence and post-generation checking.

A

Images only

Captioner views T1 and T2

BASELINE

B

+ predicted mask

Add evidence produced by the detector

PROPOSED INPUT

C

+ ground-truth mask

Oracle evidence; upper bound only

ORACLE CONTROL

D

+ consistency check

Validate B's claims against predicted evidence

PROPOSED FULL

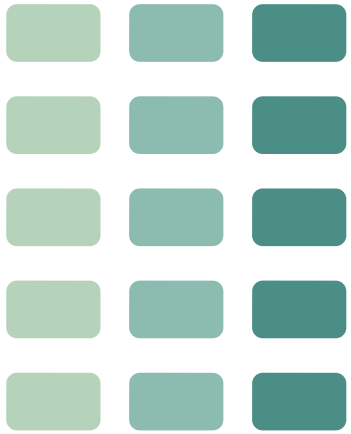
Question: Do explicit change events improve factual accuracy without materially weakening text quality?

13 / EVIDENCE STANDARD

Measure robustness. Audit factual claims.

Use the standard metrics for comparability - and independent human review for claim accuracy.

ROBUSTNESS MATRIX



5 perturbations × 3 severity levels

BLINDED HUMAN FACTUAL AUDIT

≥ 100 LEVIR-MCI test pairs

2	blind annotators / caption
3	claim labels: supported / unsupported / contradicted
κ	inter-annotator agreement

14 / WHAT SUCCESS WOULD LOOK LIKE

Targets, not promises.

We will publish negative or null findings if the evidence does not support the hypothesis.

O1

≤ 2 F1 pts

from published detection baselines

O2

≥ 25%

smaller F1 drop under shift

O3

+10 pp

supported-claim precision vs image-only

O4

< 5 s

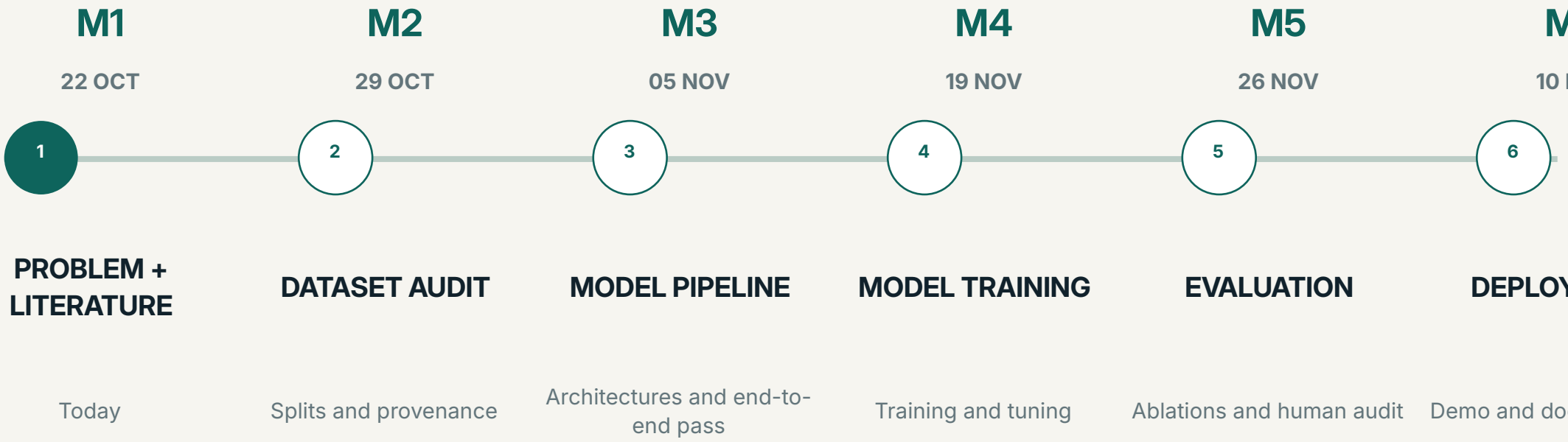
per 256×256 pair on Colab T4

All numeric goals are proposed acceptance criteria in the team report; no experiment has yet established them.

15 / ON THE CALENDAR

A six-milestone path to a defensible demo.

Milestone 1 closes the problem definition and research review; implementation follows.



Weekly technical review • written milestone report • presentation • code • individual WORKLOG

THE PROJECT THESIS

FROM PIXELS TO PROOF.

> Not another change map.
> A clearer claim - with visible evidence.



TERRAWATCH / GROUP 7

Milestone 1 • IIT Madras BS

APPENDIX

Research foundations & source trail.

Full citations, dataset provenance and related work are in the Milestone 1 report.

01 **Daudt et al., 2018**
Fully Convolutional Siamese Networks

02 **Chen, Qi & Shi, 2021**
Remote Sensing Image Change Detection with Transformers (BIT)

03 **Bandara & Patel, 2022**
ChangeFormer / transformer-based Siamese network

04 **Liu et al., 2022**
LEVIR-CC & RSICCformer

05 **Liu et al., 2024**
Change-Agent & LEVIR-MCI

06 **Corley et al., 2024**
A Change Detection Reality Check

07 **Deng et al., 2024**
ChangeChat

08 **Ricci et al., 2026**
Change captioning meets LLMs and VLMs

Official GitHub PR #10: github.com/shoaib2000857/Group-7-DS-and-AI-Lab-Project/pull/10